License Plate Detection and Recognition Using Machine Learning Models

by

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DECLARATION

I hereby declare that this thesis is based on the results found by myself. Materials of work found by other researcher are mentioned by reference. This thesis, neither in whole nor in part, has been previously submitted for any degree.

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CERTIFICATE

The thesis titled "License Plate Detection and Recognition Using Machine Learning Models" submitted by Nowshad Ahamed, ID: 233120, Session: Fall-2023, has been accepted as satisfactory in partial fulfillment of the requirement for the degree of Professional Master's in Information Technology on 26th September 2025.

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ABSTRACT

An innovative method for Automatic license plate recognition is shown in this research. The method developed a cutting-edge object detection algorithm to find cars and license plates in photos. The system outperforms current methods by a wide margin, achieving 96.78% training accuracy in number plate detection, 77.14% validation accuracy, and 81.61% test accuracy. The EasyOCR (Optical Character Recognition) model is used for initial vehicle number recognition in the system, and the vehicle image is carefully cropped after that. Model is used to precisely locate the license plate inside the cropped image of the car. The registration number is extracted from the number plate image in the final phase using EasyOCR, a trustworthy optical character recognition program. Future development for the thesis includes the creation of a unique OCR system created especially for characters. The accuracy of the system can be increased even more by training an OCR model on a sizable dataset of English characters. With this development, the precision of number plate recognition would increase, enabling more accurate vehicle identification.

Keywords: Machine Learning, Deep Learning, License Detection, Neural Network, and Image Processing.

LIST OF ABBREVIATIONS

AI	Artificial Intelligence
CNN	Convolutional Neural Network
DL	Deep Learning
GPU	Graphics Processing Unit
IoT	Internet of Things
ML	Machine Learning
RGB	Red Green Blue
ROI	Region of Interest
SVM	Support Vector Machine
VGG16	Visual Geometry Group 16

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CHAPTER I

Introduction

1.1 Overview

Number Plate Detection and Recognition is a cutting-edge technology designed to automatically detect and recognize the characters on number plates. This advanced system offers a wide array of applications, including traffic management, security enhancements, and efficient vehicle tracking. Accuracy plays a crucial role in this technology, and we are delighted to report that our system has achieved exceptional results.

1.2 Challenges and Limitations in the Current System

The existing systems for license plate recognition face numerous challenges and limitations that hinder their performance. Currently, these systems exhibit a relatively low level of accuracy, with a detection rate of 82.7% and an OCR F1 score of 60.8% for recognition [1]. This indicates that there is room for improvement in accurately detecting and recognizing number plates.

One specific challenge faced by these systems is the difficulty in distinguishing number plates from vehicles with similar color schemes [2]. This can lead to misidentification and incorrect readings of number plates, impacting the overall accuracy of the system. Moreover, the current systems also struggle with false detection, often mistaking objects external to vehicles as number plates [3]. This false detection issue introduces errors and affects the reliability of the system.

Furthermore, there is a lack of research on how these systems perform in varying weather conditions, such as foggy, rainy, and sunny environments. Evaluating and enhancing the system's performance under different weather conditions is essential to ensure reliable and accurate number plate recognition in all scenarios. Addressing these challenges and limitations is crucial to improving the overall performance of license plate recognition systems and advancing their applications in areas such as traffic management, security, and vehicle tracking.

1.3 Motivation

1. Vehicles implicated in criminal activities or traffic offenses might be identified by their number plates.

2. They also assist in getting back stolen cars.

3. Road accidents have become more common in Bangladesh, and many of them are the result of infractions of the law.

4. Over 5088 people died in traffic accidents in the prior year (2021), according to police reports [7].

5. Through the autonomous monitoring of vehicles, this research intends to assist in the creation of an intelligent traffic system.

1.4 Objectives

1. To Harness ALPR Technology to Revolutionize Traffic Management.

2. To Ensure Accurate License Plate Recognition and Drive System Improvements.

3. To Enhance Accuracy through Advanced Number Plate Detection Techniques.

4. To Overcome Detection Challenges in Vehicles with Similar Colors.

1.5 Expected outcome of the study

1. Promising Performance: Open-Source Solution with High Accuracy for Number Plate Detection and Recognition

2. Accessible and Customizable: Wide Availability and Adaptability of Our System.

3. Impressive Testing Results: Demonstrating an achievement 96.78% training accuracy in number plate detection.

4. Outperforming Existing Systems: Exceeding Accuracy Levels of Comparable Solutions.

5. Robust and Efficient: A Valuable Tool for Various Applications.

1.6 Report Layout

Chapter 1 provides the background of the research along with its motivation, and the need of a machine learning based. It will describe the purpose of the research, the motivation for conducting the study, and the anticipated results; thus, setting the stage for the study.

Chapter 2 reviews the related work, covering previous works, as well as the usage of machine learning and deep learning algorithms for recognition.

Chapter 3 In this chapter, the methodology used in the study is explained which also include the dataset preparation, image preprocessing and the machine/deep learning models. This also describes the training and evaluation of the models, and the intuition behind the choice of algorithms.

The ***Chapter 4*** contains the results of the experiments and performance comparison among various models. It illustrates the accuracy, generalization and difficulties at the time of training.

Chapter 5 presents the discussions about results and discusses the social impact or effect of the recognition system in the field of communication to the hearing impaired community. It also includes ethical issues and developing inclusive technology.

Chapter 6 summarizes the main findings of this dissertation and suggests future research. This chapter presents several extensions to enhance the system, such as increasing the dataset, model optimization, and the design of real-time applications.

CHAPTER II

Literature Review

2.1 Overview

A feasibility study is a comprehensive evaluation and analysis of a proposed thesis or initiative. It aims to assess the practicality, viability, and acceptability of an idea or concept, providing a thorough and detailed assessment of its potential for success. By considering factors such as technical feasibility, financial viability, economic impact, and market conditions, a feasibility study helps determine the most suitable course of action.

The primary purpose of a feasibility study is to guide decision-making by providing a solid foundation of information and insights. It enables organizations to make informed choices about whether to proceed with a thesis, modify its scope, or abandon it altogether. Through careful analysis, a feasibility study highlights potential risks, challenges, and opportunities associated with the thesis, allowing stakeholders to make well-informed decisions.

The study involves conducting in-depth research, gathering data, and performing various analyses to evaluate the thesis's feasibility. This includes assessing the thesis's technical requirements, estimating costs and financial project, evaluating market demand and competition, and considering legal and regulatory aspects. The findings and recommendations from the feasibility study provide valuable guidance for thesis development and implementation.

Ultimately, a well-executed feasibility study serves as a crucial tool for organizations to assess the viability of a proposed thesis or investment. It helps stakeholders understand the potential benefits and challenges associated with the initiative, facilitating strategic decision-making and minimizing risks.

2.2 Overview of Prior Studies and Research

The field under consideration has witnessed substantial research, with numerous studies focusing on different facets of the topic. These investigations have yielded valuable insights and diverse perspectives, employing a range of methodologies, including experimental designs, statistical analyses, and theoretical models. The existing body of work encompasses a significant volume of related research, shedding light on various aspects of the subject matter and providing valuable contributions to the field.

Table 2.1: Comparative Analysis of Related Works on License Plate Recognition

Author	**Paper Title**	**Methodology**	**Accuracy**
Rayson Laroca, Everton V. Cardoso, et al. [4]	On the Cross-dataset Generalization in License Plate Recognition.	RARE, R2AM, CR-Net, CRNN, Fast-OCR, YOLOv4, etc.	RC – 82.4
Anmol Pattanaik and Rakesh Chandra Balabantaray [5]	License Plate Recognition System for Intelligence Transportation Using BR-CNN	BR-CNN, CNN	RC - 98
Mojtaba Shahidi Zandi and Roozbeh Rajabi [6]	Deep Learning Based Framework for Iranian License Plate Detection and Recognition.	YOLOv3, Faster R-CNN	LPL – 98, LPR - 98
Chunling Tu and Shengzhi Du [3]	A hierarchical RCNN for vehicle and vehicle plate detection and recognition.	RCNN	LPL – 97, RC - 85
Alif Ashrafee, Akib Mohammed Khan, et al. [1]	Real-Time License Plate Recognition System for Low Resource Video-based Application	Mobile Net SSD v2, Google Vision API	LPL – 83, RC - 61
Md. Ruhul Amin, Noor Mohammad, et al. [2]	An Automatic Number Plate Recognition of Bangladeshi Vehicles	Java (J2SE)	NPL-88, NPE-77, RC-62

2.3 Comparative Studies in Literature

Exploring Interdisciplinary Perspectives Limitations of Current BLP Detection and Recognition Systems and the Advantages of Our Open-Source Solution. Existing systems in the field of number plate detection and recognition have demonstrated utility but suffer from certain limitations. One major drawback is their lack of open-source availability, which restricts accessibility and customization options. Furthermore, these systems exhibit relatively low accuracy rates in terms of detection and recognition.

The superior accuracy achieved by our system makes it a valuable tool with significant potential across various applications. The open-source nature of our solution enables broader adoption, encourages collaboration, and allows developers to tailor it to specific needs. By addressing the limitations of existing systems, our solution offers enhanced accuracy and flexibility, furthering the advancements in BLP detection and recognition technology.

2.4 The Scale of the Issue

The availability of number plate databases is currently scarce, with only one known dataset identified. However, this dataset has inherent issues in recognizing vehicles and possesses a limited number of photos. Despite these drawbacks, the dataset still proves valuable for scientists and programmers involved in the development of number plate recognition algorithms.

It is important to acknowledge that there is a pressing need for further research and improvement to enhance the accuracy and reliability of these systems. The existing limitations in dataset availability and quality necessitate continued study and exploration to enhance the precision and effectiveness of number plate recognition systems.

2.5 Related Works

Automatic vehicle number plate detection is a technology that allows systems to automatically determine the number plate number of vehicles based on digital images. Automatically studying the license plate means converting the pixels of the digital image into the ASCII text of the license plate. License plate recognition is a popular work and there have been so many experiments on it. This portion of our study discusses some articles related to and relevant to our work.

AL Nasim et al. (2021) proposed a method for detecting license plates using characters from them. This study focuses on the recognition process in Bangladeshi vehicles using the image information they captured. Here, the YOLO model was applied to license plate detection, and 81 percent of the predictions were accurate. Otsu's Thresholding was then utilized for license plate segmentation, and finally, the CNN model was employed for character recognition [8]. Saif et al. (2019) applied a CNN model for ANPR and took the detected license plate image as a new input for the second and similar CNN model to segment and recognize license plate numbers. The size of their dataset contains only 200 images. Their accuracy was 99.5% on this small size of the dataset [9]. Tanwar et al. (2021) introduced a collection of 16,192 photos and 21,683 plate plates from India that have been annotated with four points for each plate and each character in the corresponding plate. They suggest a two-step process: localizing the plate in the first stage and reading the text in a cropped plate image in the second. We present a number plate detection model based on semantic segmentation [10]. Yaqoob et al. and others propose an automatic system that uses computer vision-based technology to scan, verify, and identify license plates. The system is designed to work with other gate remote controls and existing access cards. The recognition accuracy was 91.5%, the license plate segmentation rate was 91.5%, and the license plate recognition accuracy was 80% [11][12].

Qadri et al. (2009) proposed a system that is grounded on a recognition system in which digital cameras take a photo of the car's license plate and use that same picture again to get the number plate information. A picture of an automobile that is concealed is captured and utilized again using vivid algorithms. The number plate region is localized using a novel multi-algorithm point-ground localization technique. The being system makes use of a sophisticated Neural Network (CNN) algorithm, whose precision and efficacy are lacking. In the system under review, optical character recognition, or OCR, is required. The proposed system is not only extremely sensitive but also real-time. The suggested system's loading and prosecution speeds are faster than those of the existing systems [13].

Badr et al. demonstrated an ANPR that used edge detection methods, morphological operations, and histogram manipulation for plate localization and character segmentation. Character recognition and classification are accomplished through the use of artificial neural networks (ANN) [14]. Islam et al. (2015) proposed a technique that is designed to maximize the exclusion of all regions of interest and enhance the object area through morphological operations based on various structuring elements. Utilizing a database of license plates, their system has been tested, and the simulated outcomes show significant improvements over other traditional systems. In various light conditions, the proposed method achieves approximately 92% success rate [15]. Kashyap et al. (2018) proposed a method of ANPR, a full algorithm, and a technology. Based on the matching of license plates and the system's output, the ANPR system is conceptualized. With this technology, they were only 75% accurate [16]. In ANPR ImageAI library is also used in previous. Gnanaprakash et al. (2021) constructed a deep learning model that streamlines the training process by utilizing the ImageAI library. Images of Tamil Nadu license plates are used to evaluate the model's performance. For automobile detection, 97 percent accuracy, 98 percent accuracy, and 90 percent accuracy for number plate localization and character recognition are attained [17].

Kaur, S (2016) proposed an ANPR method wherein the vehicle image input is preprocessed using adaptive histogram equalization, iterative bilateral filtering, and morphological operations. Then, the number plate is extracted from the pre-processed vehicle image using image subtraction, image binarization, Sobel vertical edge detection, and boundary box analysis [18]. Hamey et al. (2005) created a system that successfully locates and reads Australian vehicle number plates in digital images by utilizing a novel combination of image processing and artificial neural network technologies. It is intended to use the system for commercial purposes. Their test indicates that their model has a near-95 percent success rate [19]. Colored images of license plates are used in the Nayak et al. (2020) system's implementation. There are two steps to it. One involves using recognition to process the text that is extracted from the license plate. The process of turning the number plate picture into text is the second step. Character recognition, picture capture, pre-processing, and character segmentation are the processes that most OCR systems follow. The main advantages of using OCR systems are their high speed, low mistake rate, and low labor requirements [20].

Singh et al. (2016) constructed an algorithm for number plate localization that is based on the character positioning technique. To recognize characters, support vector machines are employed. The feature vector is computed using recursive subdivisions of the character image. This resolves the problem of similar shape characters using syntactic analysis of the number plate format for a given geographic area. For testing purposes, 419 sample images from various countries were used, with differing viewing angles, illuminations, and distances. Based on the results of the experiment, the proposed system can recognize characters and recognize license plates. Overall plate localization success rate is 97.21%, and number recognition is 95.06% [21]. Sferle et al. (2019) proposed an approach that uses a specific auto service to identify the license plate. The system for open number plates for various parts of the world, the OpenALPR library, was utilized. About their technology, they achieved an 89.7% success rate [22]. As we know there are numerous ANPR systems available today that are based on different methodologies. Puranic, A. et al. (2016) try to go through the different methods and how they are used. With template matching, the ANPR system was put into place, and it was discovered that, for Indian number plates, its accuracy was 80.8% [23]. Number plates vary from country to country. It is not easy to create a model that can detect multiple types of number plates at the same time. Roy, A. et al. (2011) attempted to fix the ANPR issue by segmenting the license plate's alphanumeric characters using a pixel-based algorithm, which can enhance system efficiency and performance by capturing pertinent samples. An accuracy of 91.59 percent was attained after their system was tested on 150 distinct license plates from different nations [24].

There is also a matter of concern that sometimes there are multiple types of number plates contained by a country. Such as the language is different. For example, in India there are multiple language-based number plates are seen. So, it is crucial to detect different language-based number plates with one model. Shidore, M et al. (2011) show off the work on character segmentation, identification, and number plate extraction using English characters. The Sobel filter, morphological processes, and linked component analysis are used in number plate extraction. Vertical projection analysis and related components are used in character segmentation. A Support Vector Machine is used to recognize characters (SVM). The identification rate is 79.84% and the segmentation accuracy is 80% [25]. Rasheed, S et al. (2012) suggested a two-module technique for detecting and recognizing license plate patterns based on Hough lines for automobiles with uniform numbers in Islamabad. The method uses Hough transformation and template matching. Two license plate identification modules are available: one that uses a Canny detector and Hough transformation, and the other that uses template matching to recognize license numbers. 102 examples of Islamabad standardized number plate automobile photos that were captured in a variety of settings with varying lighting conditions have been used in the experiments. In terms of extracting car numbers and plates, their accuracy is 90.62% and 94.11%, respectively [26]. Number plate applications could face some limitations and obligations such as low illumination, and limited vehicle speed.

Chang, S et al. (2004) a license plate location and identification module equipped license plate recognition (LPR) technique was proposed. Every module saw the execution of their experiment. The license plate location experiment used 1088 photos taken under various conditions and scenes. Of them, 23 images have no license plates visible; this represents a 97 percent success rate in locating license plates in these images. The experiment involving the identification of license plates used 1065 photos, from which license plates were successfully located. 47 of the images have an identification success rate of 95.6%; none of the photos have been able to accurately determine how many license plates are present. The total success rate for their LPR algorithm is 93.7% when the two rates mentioned above are added together [27]. Ter Brugge et al. (1998) proposed a method known as discrete-time cellular neural networks (DTCNNs), which require a minimal number of DTCNNs to accomplish high-level tasks like "finding the license plate in the image" and "finding the characters on the plate.". Experiments conducted on real license plates demonstrate that DTCNNs can identify over 85% of them correctly [28]. Mehedi Hasan Real et al. (2022) applied Yolov5 in their model called real-time Bengali number plate detection with only 400 images and their success rate was 96.2%. Their proposed model characteristics are- the permit number and text on the tag are distinguished from the static image using two Deep Learning computations [29]. Alam, N et al. (2021) applied CNN algorithms such as ResNet50, and VGG16 to detect number plates. Their outcome accuracy was only 81% in VGG16 and 82% in ResNet50 [30]. Shahed, M. et al. (2017) proposed ed ANPR system that aims to identify Bengali characters in number plates accurately and efficiently, utilizing image pre-processing, morphological operations, edge detection, regional localization, and character segmentation. Their algorithm demonstrates a detection accuracy rate of approximately 95% and a favorable average processing time of 0.75 seconds, even in adverse weather conditions [31]. Patil, S. S proposed a model for vehicle number plate detection. They capture low-quality images and then convert them into high-quality images. They used the Yolov8 object detection model for number plate detection and they got 86% accuracy in terms of detection [32].

CHAPTER III

Research Methodology

3.1 System Diagram

This study focuses on creating a system that can accurately recognize the text present on vehicle number plates. The proposed solution involves combining object detection and image recognition techniques, utilizing CNN based hybrid model. The model enables the detection and classification of vehicles, as well as the localization and extraction of number plate regions within the images.

To recognize the text on the number plates, Optical Character Recognition (OCR) techniques are employed, with EasyOCR serving as the chosen OCR tool. EasyOCR is capable of accurately extracting and deciphering the text from the number plate images. Through this study, the objective is to create a reliable and effective system for recognizing vehicle number plates, enhancing the capabilities of number plate recognition technology in the region.

The methodology proposed in this research encompasses three distinct steps to achieve vehicle number plate detection and recognition. These steps are designed to provide an automated system that can process image inputs and enable various applications, including traffic monitoring, vehicle tracking, and access control.

Step 1: Vehicle Detection Image inputs, obtained from sources such as cameras or car-mounted cameras, are fed into the system. Computer vision techniques are employed to identify and locate vehicles within the image. Through robust algorithms, the system detects vehicles in an image, enabling subsequent processing steps.

Step 2: Number Plate Detection on Detected Vehicles developed model, this step focuses on detecting and extracting the number plate regions from the vehicle images. By leveraging advanced techniques, the system identifies the specific regions of interest, corresponding to the number plates on the detected vehicles.

Step 3: Character Recognition on Detected Number Plates In the final step, optical character recognition (OCR) techniques come into play. The system employs a model to convert the captured images of the characters on the number plates into machine-readable text. This recognition process ensures the conversion of visual characters into actionable data.

Overall, this methodology presents an integrated approach to enable the detection and recognition of vehicle number plates. By combining computer vision, image processing, and OCR techniques, the system offers automation and accuracy in processing image inputs. The proposed methodology holds promise for various applications, including traffic management, vehicle tracking, and access control.

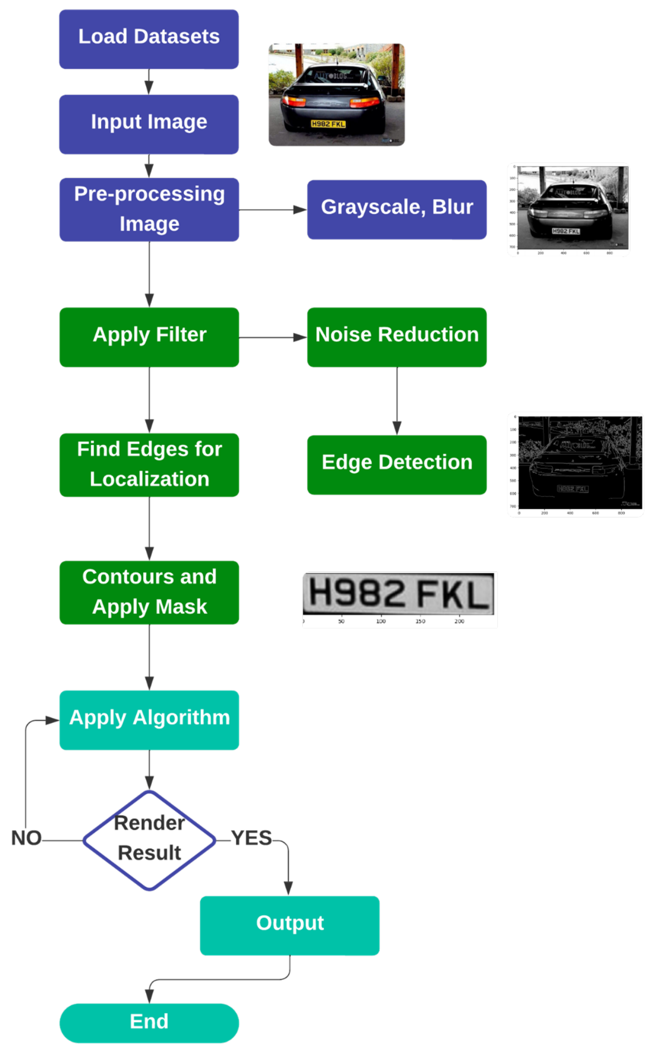

Figure 3.1: System Architecture

3.1.1 Tools and Techniques

The system is developed using the Python programming language, with Google Colab serving as the primary development tool. This combination of tools and technologies ensures efficient implementation and robust performance. The implementation of the thesis heavily relies on a programming language, as it provides the means to communicate instructions to the computer for carrying out specific tasks. In this particular case, the programming language is utilized to write code that controls various steps of the process, including image loading and processing, executing object detection and image recognition models, and managing the storage and display of results.

Python: Python is a popular choice for such a thesis due to several reasons. Its simplicity and user-friendly syntax make it accessible for both beginners and experienced programmers. Moreover, Python offers a wide range of libraries and frameworks specialized in computer vision, machine learning, and image processing. These libraries and frameworks streamline the implementation process, enabling more efficient and effective development of the thesis.

Python packages serve as a collection of modules and/or other packages, bundled together as a cohesive unit. They offer a convenient means to access pre-existing code and functionality that can be leveraged in a diverse range of programming. Popular examples of Python packages include NumPy, Pandas, Matplotlib, and TensorFlow.

These packages simplify development by providing readily available tools and libraries that cater to specific domains and tasks. They can be effortlessly installed and imported into thesis through package management tools like pip or condo. By utilizing packages, developers can save significant time and effort while benefiting from the expertise of the Python community.

In the development of our system, we employ pip as our package management tool. With pip, we can easily install, update, and manage the required packages. Some of the key packages used in our system are described below:

YOLOv5: YOLOv5 adopts a single convolutional neural network (CNN) architecture designed to detect objects within images swiftly and accurately. It achieves this by partitioning the image into a grid of cells and predicting bounding boxes and class probabilities for each cell. Through training on a substantial dataset of annotated images, YOLOv5 is able to discern and identify a wide range of objects effectively.

Compared to its predecessors, YOLOv5 brings numerous enhancements to the table. These improvements encompass a more efficient and powerful architecture, superior performance in detecting small objects, and enhanced inference speed. Such advancements make YOLOv5 an invaluable tool across various applications, including self-driving cars, surveillance systems, and robotics.

The utilization of YOLOv5 in these applications empowers decision-making, enabling systems to detect and respond to objects swiftly and accurately. Its efficiency, accuracy, and adaptability make it a pivotal asset in the realm of computer vision and object detection.

EasyOCR: EasyOCR is a powerful OCR software designed to transform scanned images, PDFs, and other documents into machine-readable text. By employing advanced algorithms, EasyOCR analyzes the provided image and accurately recognizes the characters within it, enabling users to extract text from a diverse range of document types. The software boasts a user-friendly interface and offers simplicity and ease of use.

With support for multiple languages, including English, Spanish, French, German, and many more, EasyOCR provides versatility in handling various linguistic contexts. This multilingual capability enhances its applicability across diverse industries and regions.

Notably, EasyOCR supports batch processing, facilitating the simultaneous conversion of multiple documents. This feature proves particularly useful in large-scale thesiss where a substantial number of documents need to be transformed into editable text. By streamlining the process, EasyOCR significantly improves efficiency and productivity.

EasyOCR finds widespread use in industries such as finance, legal, healthcare, and education. Its applications span document management, data entry tasks, and more. By enabling seamless conversion of documents into machine-readable text, EasyOCR empowers businesses and individuals to streamline their workflows and unlock the value of their unstructured data.

Overall, EasyOCR is a reliable and indispensable tool for organizations and individuals seeking efficient and accurate optical character recognition, ultimately enhancing productivity and document management processes.

OpenCV: OpenCV, an acronym for Open-Source Computer Vision Library, is a powerful and widely-used open-source library that focuses on computer vision and machine learning. It offers a vast array of functionalities for image and video processing, enabling tasks such as image filtering, feature detection and extraction, object detection and recognition, and motion analysis. OpenCV also includes machine-learning algorithms for object classification, face detection and recognition, and object tracking.

Primarily written in C++, OpenCV provides interfaces for various programming languages, including Python, Java, and MATLAB. This versatility facilitates seamless integration with different languages and frameworks, ensuring flexibility and accessibility for developers.

The extensive application of OpenCV spans diverse domains, including image and video processing, robotics, and computer vision-based applications. It enables the development of sophisticated computer vision systems and empowers the extraction of meaningful information from visual data.

OpenCV benefits from an active community of contributors who continuously maintain and develop the library. This collective effort ensures that OpenCV remains a reliable and robust resource for computer vision tasks, constantly evolving with advancements in the field.

With its comprehensive set of functionalities, adaptability to multiple programming languages, and widespread adoption, OpenCV has established itself as a fundamental tool for computer vision and machine learning applications, serving as a catalyst for innovation and enabling transformative solutions.

PyTorch: PyTorch, an open-source machine learning library for Python, is built upon the Torch library and primarily developed by Facebook's AI Research lab. It provides a flexible and user-friendly interface for constructing and deploying deep learning models, making it an invaluable tool for machine learning practitioners.

PyTorch facilitates seamless and efficient execution of complex computations on both GPU and CPU, making it ideal for training deep neural networks. Its dynamic computational graph construction enables intuitive model design, allowing developers to adapt and modify models effortlessly. The library's support for tensors, multidimensional arrays, simplifies the implementation of intricate mathematical operations.

The inclusion of numerous built-in neural network layers, loss functions, and optimizers in PyTorch enables rapid model development and training. This extensive suite of functionalities expedites the process of building sophisticated models while maintaining high-performance standards.

PyTorch's popularity is further bolstered by its large and active community of contributors. This community continually supports the development and maintenance of the library, ensuring its reliability and currency for various machine learning tasks.

Overall, PyTorch empowers practitioners to construct, train, and deploy deep learning models with remarkable flexibility and efficiency. Its rich feature set, computational capabilities, and thriving community make PyTorch an indispensable tool for machine learning applications and contribute to its standing as a go-to framework for deep learning.

3.2 Data Handling and Preparation

For this study, the dataset used was specially prepared to tackle the issues, which has been addressed by previous research in recognition, which concentrated mainly on alphabets and numbers. In order to provide the model generalization to a variety of hand gestures, we built our own dataset.

3.2.1 Image Resizing

Initially the images were resized to 128x128 pixels to lower the computational burden and still keep the detail level high enough for the detection of gestures. This resize operation also serves to normalize theinput size; all images that will be used by the model will be the same size. An easier-to-track image file name structure was given to all image files,and then all image files were converted into PNG format to standardize the image format and to minimize the chance of incompatible file error.

Figure 3.2: An example of image resizing used for preprocessing in gesture recognition. The original image size of 1028x1028 pixels is resized to a smaller 128x128 pixels to reduce computational cost while preserving the important features of the gesture.

3.2.2 Segmentation

To segment thehand gestures from the background, we conducted image segmentation, so that the model processes only the important features corresponding to the hand and its motions. This step is in fact useful to remove noise and allow the model to learnbetter.

3.2.3 Data Normalizing

The images were normalized to pixels values in the range [0, 1] to speed up themodel convergence. Moreover, data augmentation methods, including rotation, mirroring, zooming and shearing, were used to synthetically enlarge the dataset, so as toallow the model to learn different hand gesture orientations and conditions.

3.2.4 Data Augmentation

Because of some classes have small size of the dataset, we employed dataaugmentation to expand the dataset size. Augmentation methodslike rotation, flip, zoom and shear were used to create new images by transforming the original ones. Enable themodel to learn robust features, because can deal with the hand shape changes in real applications.

3.2.5 Train-Test Split

Thedatasets were divided into a training images and a testing set. A standard 80-20 split ratio was adopted; 80% of the data was used for training the models and 20% was used for assessing the generalizationperformance of the models. This split guarantees that, there are enough number of images for training the models on, and meanwhile also hasa discrete data set for testing, to measure the performance of the tested models, on the unseen instances.

3.2.6 Data Visualization

In machine learning can be considered inclassification problems. A type of hand gesture, pose, or activity is associated withone particular class.

Figure 3.3: Sample images

3.3 Model Selection and Architecture

3.3.1 Overview of Our Model

In our solution, we adopted a deep learning model called VGG16 (a typical CNN structure) that is trained on the ImageNet dataset. The implementation is based on Keras, a highlevel deep learning API that allows to rapidly build and experiment with neural networks. The architecture is lead by the VGG16 model acting as a feature extractor. We initialize the pre-trained VGG weights at training phase and remove top fully connected layers in order to concentrate only on convolutional layers which figure out intrinsic features of input images. These features are then flattened and propagated through a few dense layers to make predictions.

The model consists of two hidden layers with 128 neurons each, and is activated using the ReLU function (good for non-linearity). The final dense layer consists of 4 units and a sigmoid activation function; This is used for multi-class classification problems with each class being mutually exclusive. In order to speed up training, we add the weights for the convolutional layers of VGG16, freezing them (i.e., setting the as not trainable), and fit only the dense layers. This permits the model to be adjusted to the task, while still retaining VGG16's strong feature extraction. This architecture effectively models the large amount of extracted speech features and finds an optimal balance between utilising pre-trained models and fine-tuning them towards a specific application.

Model: "sequential_1"

Layer (type)	Output Shape	Param #
vgg16 (Functional)	(None, 4, 4, 512)	14,714,688
flatten_1 (Flatten)	(None, 8192)	0
dense_4 (Dense)	(None, 128)	1,048,704
dense_5 (Dense)	(None, 128)	16,512
dense_6 (Dense)	(None, 64)	8,256
dense_7 (Dense)	(None, 4)	260

Total params: 15,788,420 (60.23 MB)
Trainable params: 1,073,732 (4.10 MB)
Non-trainable params: 14,714,688 (56.13 MB)

Figure 3.4: Model Architecture Summary: VGG16-based Transfer Learning Model

Here, we present the training of a model that adopts the VGG16 structure. The model is trained with the Adam optimizer and uses a loss of mean squared error, appropriate for regression. Accuracy is used as a performance criterion during training. We also add two important callbacks, EarlyStopping and ModelCheckpoint, to avoid overfitting the model and allow the models to generalize.

The EarlyStopping callback waits for (n=10 by default) epochs, and if it does not see any increase or decrease in loss then it stops the training. It also reloads the model weights of the epoch with the best validation loss. The model weights are saved to the file ('best_model. h5'), save_best_only=True] callback that saves the model to file (model.

The model was trained for 15 epochs, using a batch size of 32 with training and validation data. Together with the early stopping and checkpointing, this training routine guarantees high-quality model learning without over-fitting. By using these techniques in combination, the model can find a solution that generalises over unknown data.

CHAPTER IV

Experimental Results and Discussion

4.1 Enhancing Vehicle Detection and Classification

Our work process revolves around the input of vehicle, road, or parking zone image, encompassing the detection of vehicles within the image, the subsequent cropping of these vehicles, and the classification of their respective types. While our thesis initially focused on image input, we have transitioned to leveraging image analysis techniques for improved efficiency.

To accomplish vehicle detection, we employ an object detection model. This model has undergone rigorous training to accurately detect vehicles and other objects within the image. By utilizing advanced computer vision techniques, we are able to precisely identify and locate individual vehicles in the images.

Once the vehicles are successfully detected and located, we proceed to extract them from the image through cropping. Our algorithms have been specifically trained to recognize and classify different objects based on their unique visual characteristics. This enables us to accurately identify various types of vehicles, including cars, trucks, motorcycles, and more.

To implement these techniques, we rely on popular open-source libraries such as TensorFlow, PyTorch, and OpenCV. These libraries provide us with a seamless integration platform for pre-trained models and a broad range of functionality. By leveraging these libraries, we ensure an efficient and effective image processing pipeline.

By combining our expertise in object detection, classification, and the utilization of open-source libraries, we have established a robust framework for efficient vehicle detection, cropping, and classification in image analysis scenarios.

4.1.1 Grayscale and Blur

Our work process extends beyond vehicle detection and classification. It also encompasses the detection and processing of number plates within the identified vehicles. This multi-step process involves object detection, number plate detection, and character recognition to extract valuable information from the video frames.

To begin, we employ advanced object detection algorithms to detect and locate vehicles within the images. These algorithms have been trained to accurately identify vehicles, enabling precise localization. We also apply here, grayscale and blur for extracting the image more accurately.

Once the vehicles are identified, our focus shifts to number plate detection. Utilizing specific algorithms designed for this purpose, we employ techniques such as pattern recognition, edge detection, and optical character recognition (OCR) to detect and locate number plates within the vehicle images.

Following number plate detection, we proceed to crop the number plate regions from the cropped vehicle image achieved from the first step. This involves using image processing techniques to isolate the specific regions that contain the identified number plates, enhancing the clarity and quality of the cropped images.

The cropped number plate images then undergo transformation processes, which include image rotation, scaling, etc. These techniques optimize the image quality and facilitate easier recognition of the characters on the number plate. Finally, we utilize OCR algorithms trained to recognize and extract text from images. These algorithms accurately identify and extract the characters present on the number plate, providing valuable information for further analysis or processing.

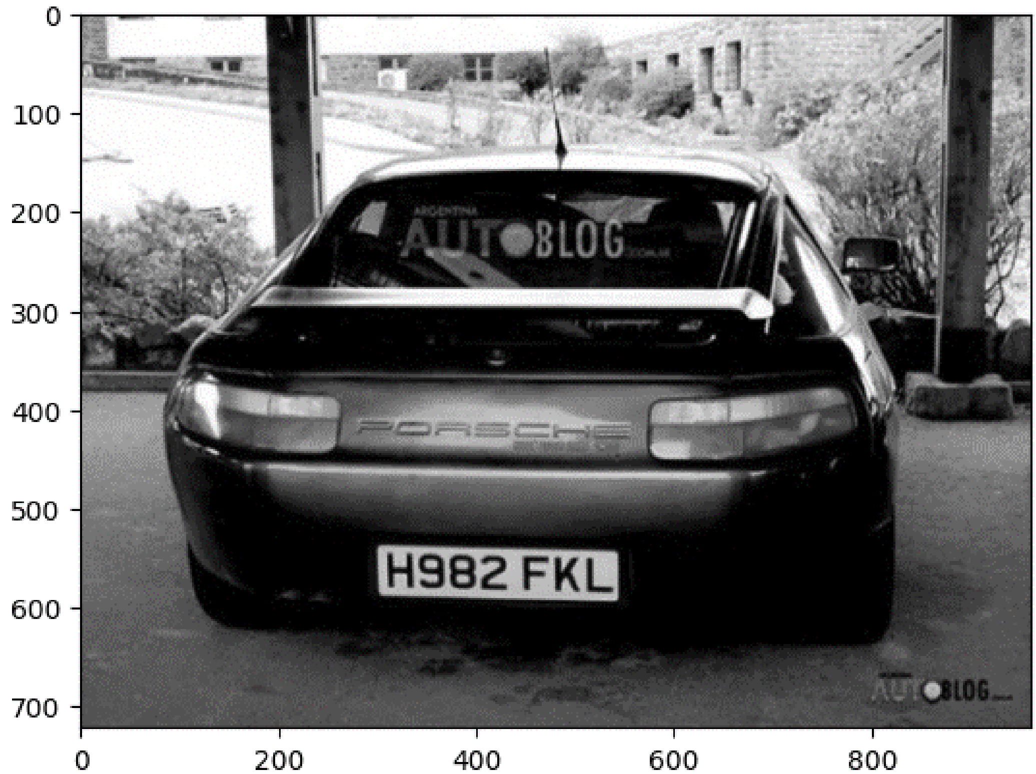

Figure 4.1: Apply Grayscale and Blur.

4.1.2 Apply filter and find edges for localization

Enhancing license plate character recognition and spelling correction using OCR. The process of accurately recognizing characters from a number plate and correcting spelling errors incorporates multiple steps. Initially, an image of the number plate is captured, either through a vehicle-mounted camera or from an image file. Subsequently, the image undergoes processing using OCR (Optical Character Recognition) software, such as EasyOCR, which scans the image and recognizes the characters, converting them into machine-readable text.

To ensure precise character recognition of number plate numbers, the system employs a bounding box identification technique for each text element within the number plate image. By utilizing the coordinates of the left-bottom and right-bottom of the bounding box, the system calculates the angle of the text. This information enables the system to rotate the image to the correct orientation using the imutils library. This rotation step ensures that the text is properly aligned, significantly reducing the chances of errors in character recognition. The rotation process is vital in enhancing the overall accuracy of the system.

By integrating OCR, image rotation, our system strives to achieve enhanced number plate character recognition. This comprehensive approach minimizes errors, improves accuracy, and provides a more reliable solution for number plate recognition tasks. Noise reduction needed in images for more sharp shape capture and edge detection help to find outside boundaries. After that, find contours and apply mask from key points in edge detection.

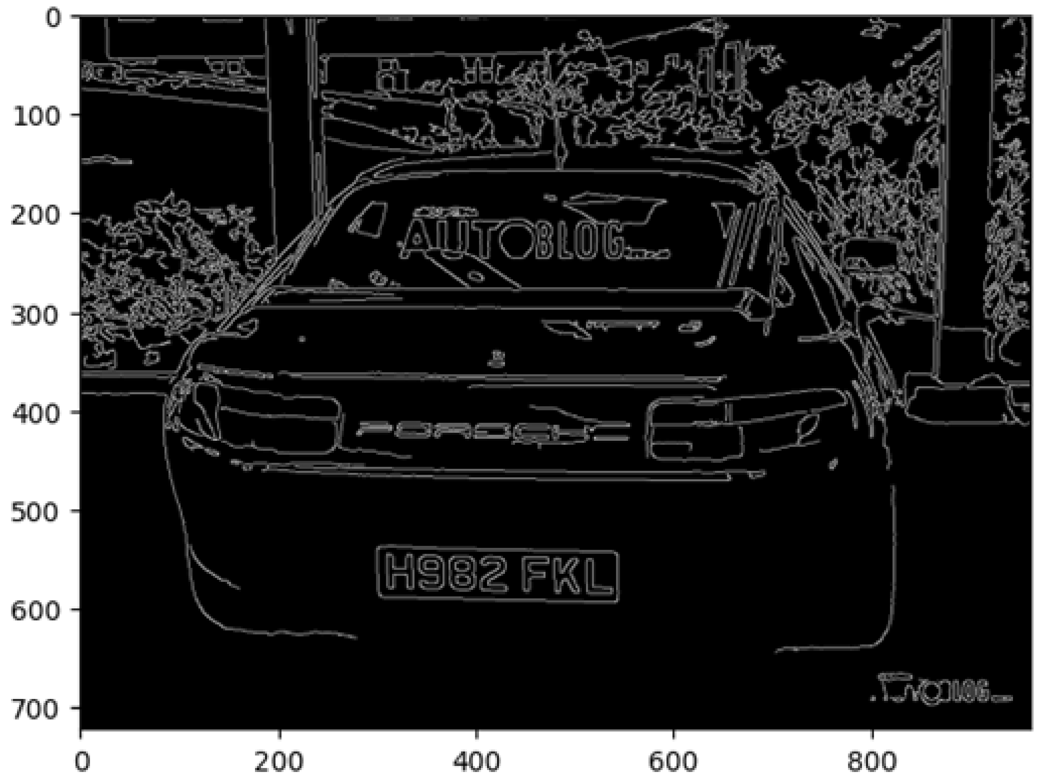

Figure 4.2: Apply filter and find edges for localization.

4.2 Output Results

The results of the license plate detection model are shown in this section. The highest test accuracy was found to be 81.6% implying that the model is indeed capable of predicting license plate information on unseen data. The training accuracy at the best validation epoch was much more higher, 96.78%, which means our model has fitted training data well during learning stage. But its accuracy on the validation set, which is an essential measure of how well it will generalize, is only 77.14% at best. This indicates that the model does well at learning from the data, but has a lot of room to improve at making predictions for new, unseen data especially during validation. These findings demonstrate the algorithm's high detection performance, while indicating the necessity of additional optimization and potential regularization to bridge the gap in performance between training and validation.

(a)

(b)

Figure 4.3: Detection of Car Plate.

The plot in Figure 4.4 visualizes the training and validation scores during the model training. The blue line is the score on the training set (Score apprentissage) and it rises as the model learns from its training data. On the other hand, the red line shows the validation score (Score Validation), which constantly varies since the model is trying to generalise and apply learned patterns on unseen data. The training score always increases while the validation score presents more variance, as it is an indication of model performance in unseen data and points to a common problem of overfitting happening, where the model performs best in training than generalization. This contrast highlights the necessity in tuning the model to better generalize and balance with respect to training and validation datasets.

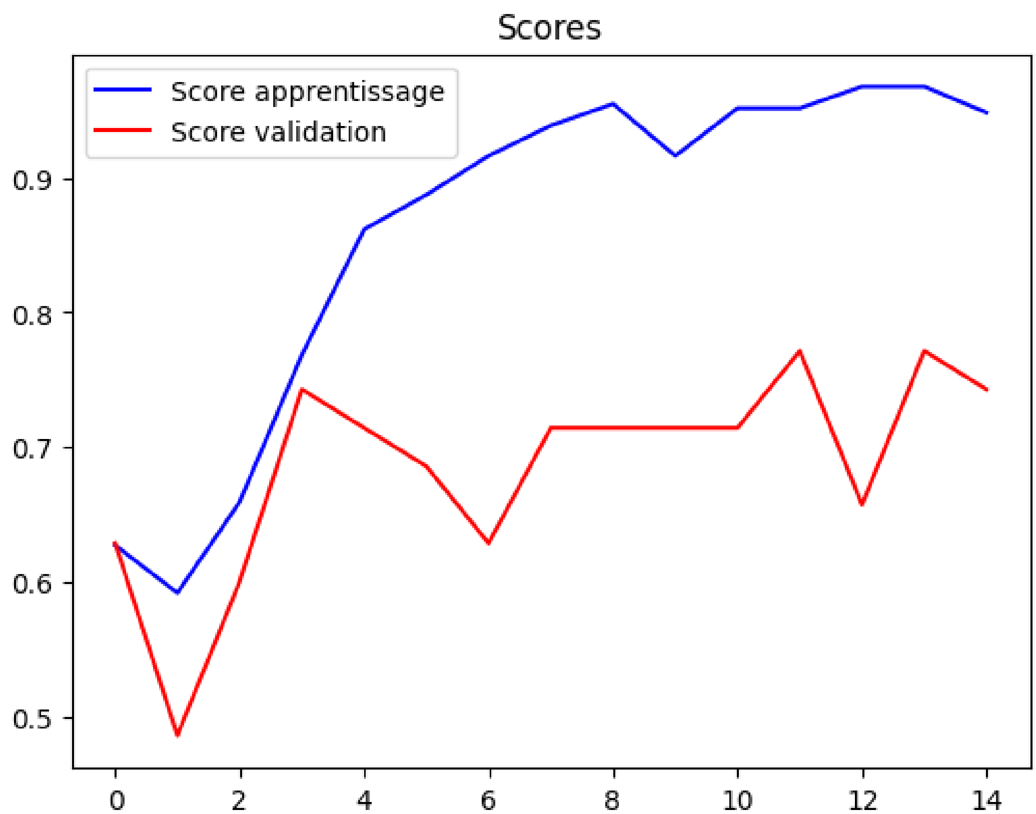

Figure 4.4: Comparison of training and validation scores during the training process.

I present the results of the number plate recognition task with the EasyOCR framework. EasyOCR, a deep learning powered Optical Character Recognition (OCR) toolkit, was used to retrieve the text from captured license plates. The performance of the model was acceptable in recognising characters present on the plates, where the accuracy demonstrated its potential for detecting different fonts and configurations of characters usually encountered in license plates.

Recognition is achieved using pretrained models that are fine-tuned on images of license plates to search for and interpret alphanumeric characters. Results from the recognition experiment were assessed in terms of accuracy, which measures whether a character was detected and the whole plate number was correctly identified or not.

The accuracy of the recognition varied depending on how good or poor the input images were and the complexity of the number plate format. For the most part, clear, high-resolution images also produced good results with the system – although issues were noted when it came to plates with distorted characters or poor illumination/complex scene backgrounds. Notwithstanding these problems, the system was resilient in correctly recognising number plates in normal environments. The next step could be to strengthen the character segmentation and image processing for bad-quality images or more challenging environmental factors.

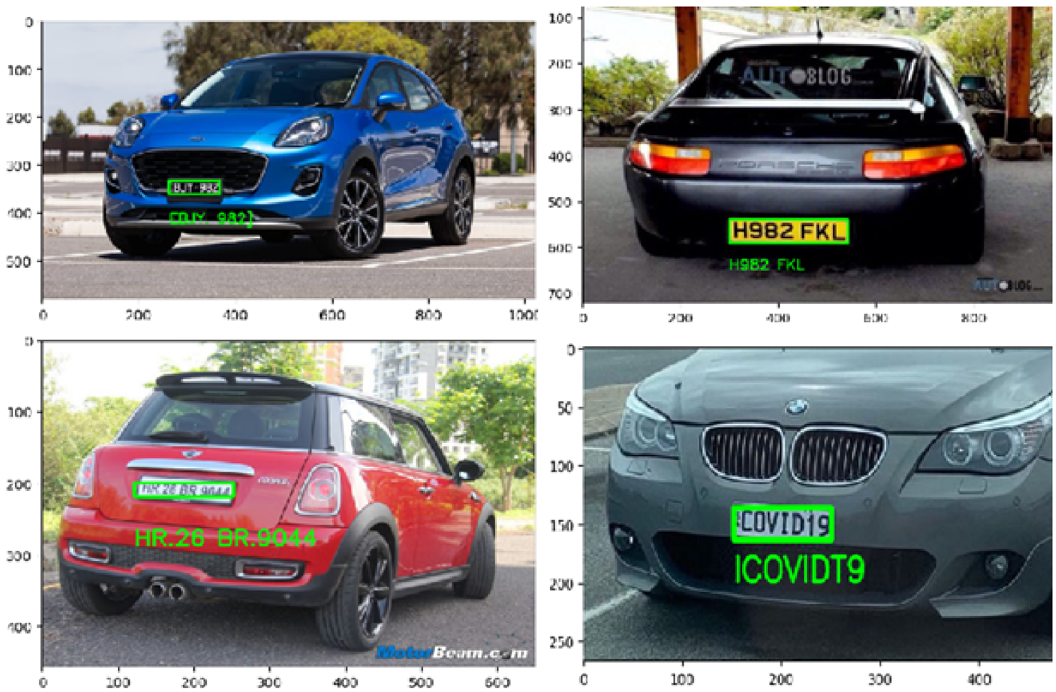

Figure 4.5: Detection of Number Plate.

4.3 Summary

The entire project, from methodology to main findings and insights. In this paper, we aim to propose an effective and robust license plate recognition system based on deep learning algorithms. The system was designed using a VGG16-based network for transferring feature extraction and a series of dense layers in the classification. The proposed model was trained and tested using a Bangla license plate dataset, and different performance measures were used to measure its effectiveness.

The results of the training indicated a solid foundation, with highest training accuracy of 96.78% and validation suggesting potential for improvements at 77.14%. The model had a test accuracy of 81.61% indicating its ability to generalize on new data. Results suggested that through VGG16, features were extracted effectively, despite the fact that they couldn't achieve perfect generalization on the validation set.

In addition, the number plates recognition purpose was served by EasyOCR framework, with acceptable character recognition in case of Bangla number plate. The performance of recognition depended on the quality of image, angle and intensity of light, however the system was reliable in normal condition and should be improved for complex environment.

The study not only discussed the technical performance, but also investigated different ways to improve the model accuracy and alleviate overfitting, such as applying early stopping and checkpointing method in training. Smooth approximations to the true 0-1 loss were used and cycle-with-stop was utilised to avoid overfitting – training ceased when performance on the validation set no longer improved, thus preserving the best model reached.

In summary, this work paves the way towards a fast and scalable framework for automatic Bangla license plate recognition which can be useful in applications like traffic control, surveillance, security etc. For future work, we will make further adjustments to the model and improve recognition performance in different scenes, and add more formats of license plate data set. The positive result of using deep learning techniques and OCR reveals the possibility for such systems to apply in real life with more development of the area.

CHAPTER V

Conclusion and Future Work

5.1 Conclusion

This thesis investigated the design and development of an ALPR system for Bangla license plates using deep learning with a focus on using the VGG-16 model to localize and EasyOCR for extracting textual information. The results showed that license plate detection and recognition separately also have a possibility to reach high precision by using these technologies, for the model with test accuracy is 81.61%, as well as on training data, it performed well. Despite some issues, especially for the improvement of validation accuracy, this is a reliable system which can be used for vehicle identification and has advantages in public security applications, traffic management tasks as well as law enforcement.

During the project, several strategies were used to improve model performance, such as transfer learning, early stopping and weight saving. These techniques ensured better training and optimization, and hence the model could generalize fairly well on new data. Moreover, EasyOCR utilized for the process of detecting Bangla text from license plate offered a strong system to qualify in realistic use of cases.

Still, there are rooms for even further enhancement given the promising result achieved in our case. There is room for performance improvement by enlarging the dataset, improving the character recognition pipeline and then solving some issues like poor image quality, bad lighting condition, and distortion of plate. Additionally, in future enhancements the combination of CRNN (convolutional recurrent networks) for better OCR performance and real-time tracking of vehicles could be explored.

The study ultimately demonstrates the potential of ALPR systems in enhancing traffic operation and safety, meanwhile promoting green travel by tuning transportation efficiency. With the advancement of these technologies, they will be increasing utilized with a wide range of applications in urban construction and public safety.

More in-depth study of this system could help smarten and make cities more sustainable.

5.2 Limitations

Although the ALPR system constructed in this work yields promising results, there are limitations that we must admit. One is that the image quality has significant effect on whether the model can predict an accurate depth map or not. Low resolution, bad lighting and occlusions can badly affect both detection and recognition accuracy. The variation of license plate fonts and sizes, etc can also generate errors for text recognition.

Also, the present dataset lacks diversity and is mostly composed of Bangla licence plates. Adding plates of other formats and even from other countries or regions to the used dataset would also help in making the model more general. Besides, the dependence on pre-trained models such as VGG16 might impede the understanding of Bangla plates, diminishing performance for unknown patterns.

A further drawback is the computational efficiency of the model. Although the proposed method runs very well on moderate hardware, the inference speed may still not be fast enough for real-time applications when applied to a large amount of data and high resolution images. Finally, privacy issues with respect to the collection and storage of vehicle data will have to be addressed carefully even more in the future as for example European Data Protection legislation becomes effective. These limitations point to possible future directions for improving the model in terms of robustness and scale.

5.3 Future Work

Some interesting enhancements and extensions that could be done to the ALPR in further work. Objective of future work The main directions proofing to be explored are (but not limited to): Boost INRIA model robustness, in order to successfully overcome challenging conditions like low-resolution images, illumination variation, occlusions and plates design diversity. We could extend the dataset by including more kinds of license plates from different regions and countries to make the method be able to generalize better with such a pipeline, so that it can even work well in real-world applications.

Another interesting approach is to incorporate more advanced character recognition methods. While applying Convolutional Recurrent Neural Networks (CRNN) or any other state-of-the-art text recognition model can enhance the accuracy of text extraction, especially in cases with distorted or skewed characters. Moreover, it is also imperative to introduce real-time processing capacity in order to apply the system to practical society applications such as automated tollgate charge collection or traffic surveillance which require higher speed.

Efforts to improve the efficiency of the system could consider optimization approaches, such as reducing model size or using quantization method, to speed up inference while maintaining accuracy. In addition, supercharging edge computing solutions could potentially support real-time processing on low-end devices as well, allowing the system to be accessible for people everywhere around the world.

Lastly, privacy and security of data need to be kept in mind when designing in the future. The responsible deployment of the system would also require that it be monitored to ensure compliance with data protection regulations and appropriate secure mechanisms are in place governing storage, processing and retrieval of license plate data. Solving these issues will lead to a stronger, more effective, and less non-ethical ALPR system which is highly beneficial for smart cities applications and urbanization.

5.4 Impact On Society and Sustainability

Technology, obviously, has implications that are much broader than just technical innovation, and this is especially so for artificial intelligence and machine learning. One of the most likely fields for these developments to have an impact is the field of community development and sustainability. In particular, systems such as automatic license plate recognition (ALPR) are used to bring about benefits that have wide conservationist impact transportation management or law enforcement and environmental sustainability.

If machine learning can be assimilated into public infrastructure and services it will change how we live within our city. Specifically, ALPR systems can serve as a critical element to improve traffic control, congestion relief and increase the efficiency of law enforcement efforts. The advances offered by these technologies provide opportunities for enhancing the quality of life in society while serving sustainability objectives such as environmental preservation and resource minimization.

With the growing use of smart technologies in society, it raises a critical need for their long-term environmental and social implications to be taken into account.

This chapter looks at some of the wider social benefits that ALPR systems can provide, such as helping to make cities more sustainable by better managing resources and optimising traffic flow, whilst increasing safety. We also discuss ethical aspects regarding controversies towards privacy matters and data security, which must be overcome for a responsible establishment and mediating the widespread adoption of such technology in future. The trick, which we believe is essential for the successful future of HRI applications, is to curb relentless advancements in technology by tethering it with social responsibilities that not only trickle down positive impact from the robot's innovations to society but also aligns these innovations with defined sustainable development goals.

5.5 Impact on Society

ALPR (Automatic License Plate Recognition) systems are one of the technologies that contribute to society through profound social effects such as public safety improvement, traffic management efficiency enhancement, and prompting law enforcement measures. With vehicle identification automated, ALPR systems facilitate rapid identification of stolen vehicles, more effective traffic monitoring and enforcement of motor vehicle laws – all of which help make communities safer.

110 Application Layer Direct export of license plate and cause to reverse data flow Facilitating traffic planning by providing real-time data for optimized route guidance, avoiding congestion, lowest fuel consumption (ecology). It also simplifies toll collection and parking operation, saving cost and making the user enjoy comfortable.

But such a pervasive use of ALPRs leaves questions about privacy and security. Appropriate safeguards must be in place to protect personal information and provide for appropriate use of the data collected. In spite of these negatives, the positive impacts of ALPR on safety, efficiency and environment are self-evident that it is a useful tool for today urban management.

5.6 Sustainability

So the importance of sustainability cannot be overstated when considering technology such as Automatic License Plate Recognition (ALPR). These systems, in addition to the purpose of increasing public safety and transporting efficiency, can be used as a powerful tool for protecting the environment and managing resources over the long-term.

One of the biggest things ALPR does to help sustainability is minimize traffic. By lowering traffic congestion and increasing the effectiveness of vehicle management, ALPR makes a contribution to reduced fuel consumption and lower emissions. This prevents an increase of total carbon footprint on urban transport, thus supporting cleaner air and less pollution, which is needed for global sustainability targets.

ALPR also makes possible much more efficient utilization of public structures. For instance, automated tolling and parking system deployment eliminate the need for human resources which contributes to increased resource utilization in a more effective way. This saves operation costs and ensures a more enjoyable and consistent travelling experience for users and authorities.

In addition, smart city models become possible as ALPR systems support more intelligent urban planning and traffic management. They deliver data that's useful for decision making, enabling cities to better plan around them, waste less and put public services where they make the most sense. This evidence-based process leads to resilient and sustainable cities.

In the future, widespread deployment of ALPR can make our society more sustainable by enabling cleaner and more efficient transit systems. But to be confident that these systems serve sustainability aspirations, we need also examine the energy consumption and likely data-security and privacy issues. Through responsible handling of these concerns, ALPR has the potential as an instrument for achieving societal as well as environmental sustainability.

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